



MEASLES-RUBELLA

Bulletin



An EPI Bangladesh Initiative

Vol # 2 , July 2021

Initiative supported by WHO Bangladesh

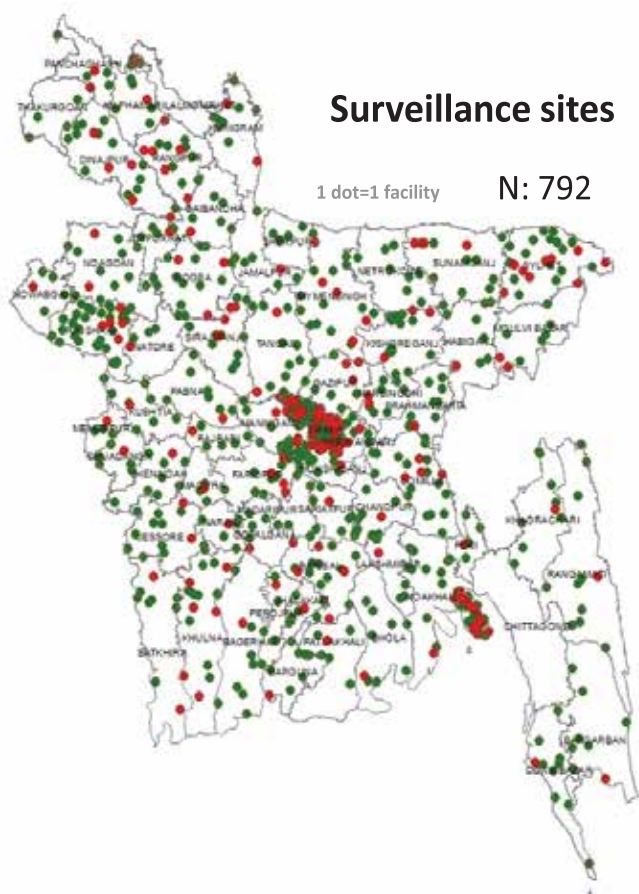


MEASLES-RUBELLA

Bulletin

Key Highlights

- 1979 : Bangladesh introduced first dose of measles vaccine
- 2003 : Laboratory supported measles surveillance initiated
- 2005/2006, 2010: Conducted nationwide MCV catchup and follow-up campaign
- 2012 : MR second dose introduced
- 2014: Nationwide MR catchup campaign
- 2015 : National Verification Committee (NVC) for measles, rubella & CRS elimination formed
- 2017 : Strategic Plan for the Elimination of Measles, Rubella and CRS in Bangladesh
- 2018 : Rubella & CRS controlled certification
- 2020: The WHO SEAR Region has adopted the goal of “measles and rubella elimination by 2023”
<https://apps.who.int/iris/bitstream/handle/10665/327923/sea-rc72-r3-eng.pdf?sequence=1&isAllowed=y>
- 2020 : Significant impact of COVID-19 on measles and rubella activities
- 2020: MR follow-up campaign conducted from 12 December 2020 -08 February 2021



Bangladesh : Surveillance sites

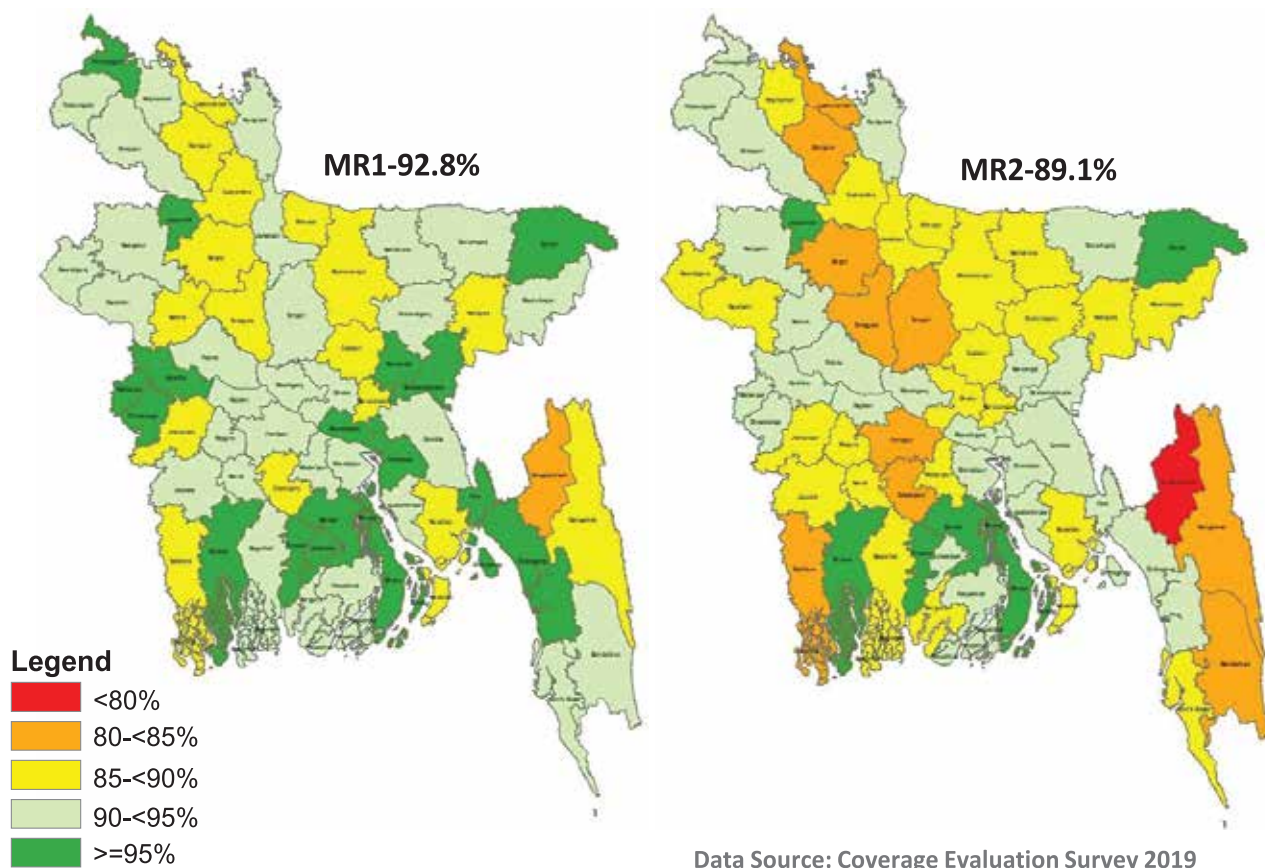
Total Active surveillance sites 167

Total Passive surveillance sites* 792

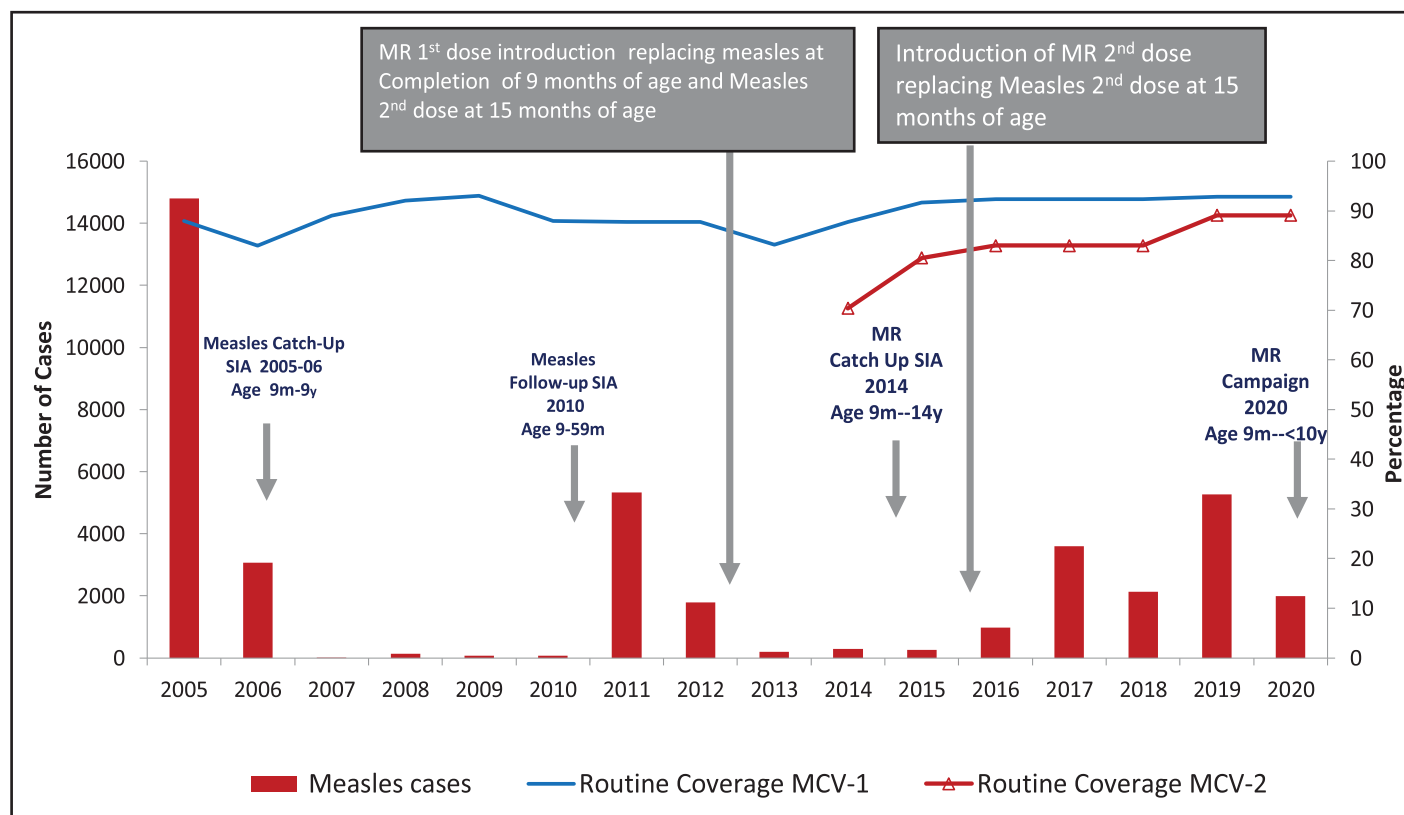
* Also Includes active surveillance sites.

- Active Surveillance Sites
- Passive Surveillance Sites

CES 2019 : Valid MR1 & MR2 vaccination coverage by age of 23 months

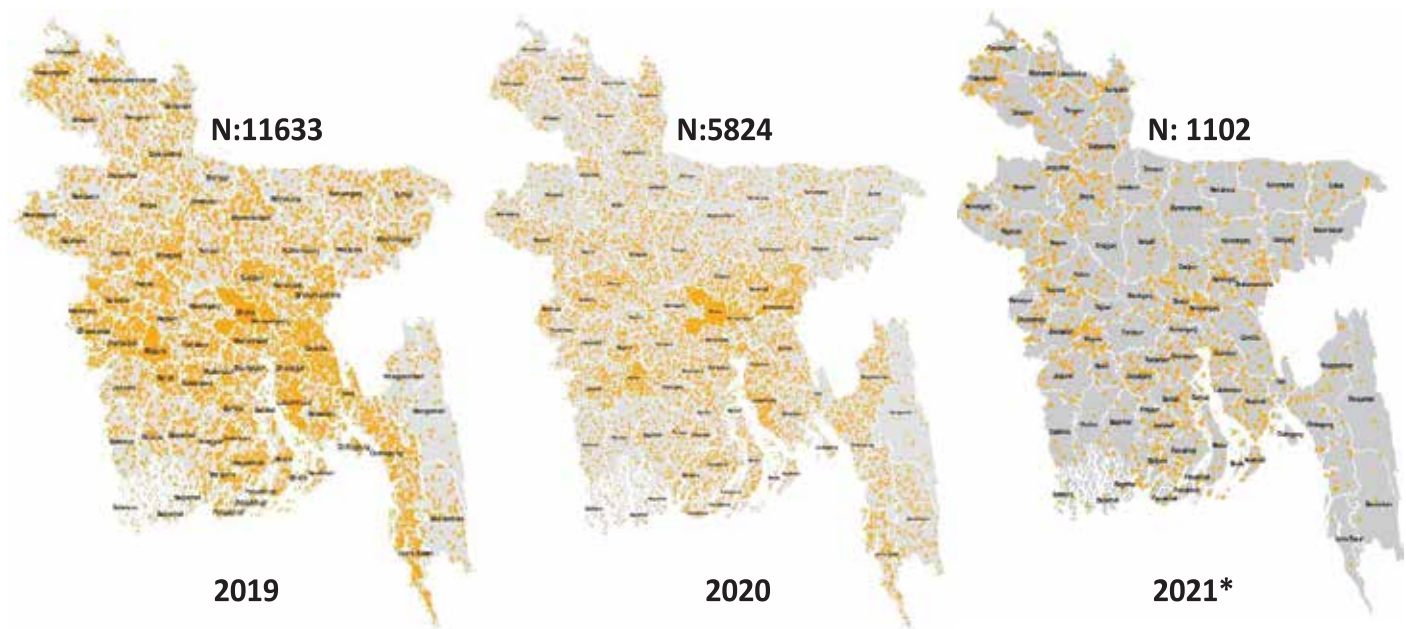


Impact of routine and supplementary vaccination on reporting of measles cases, 2005-2020



Distribution of Suspected Measles Cases 2019, 2020 & 2021*

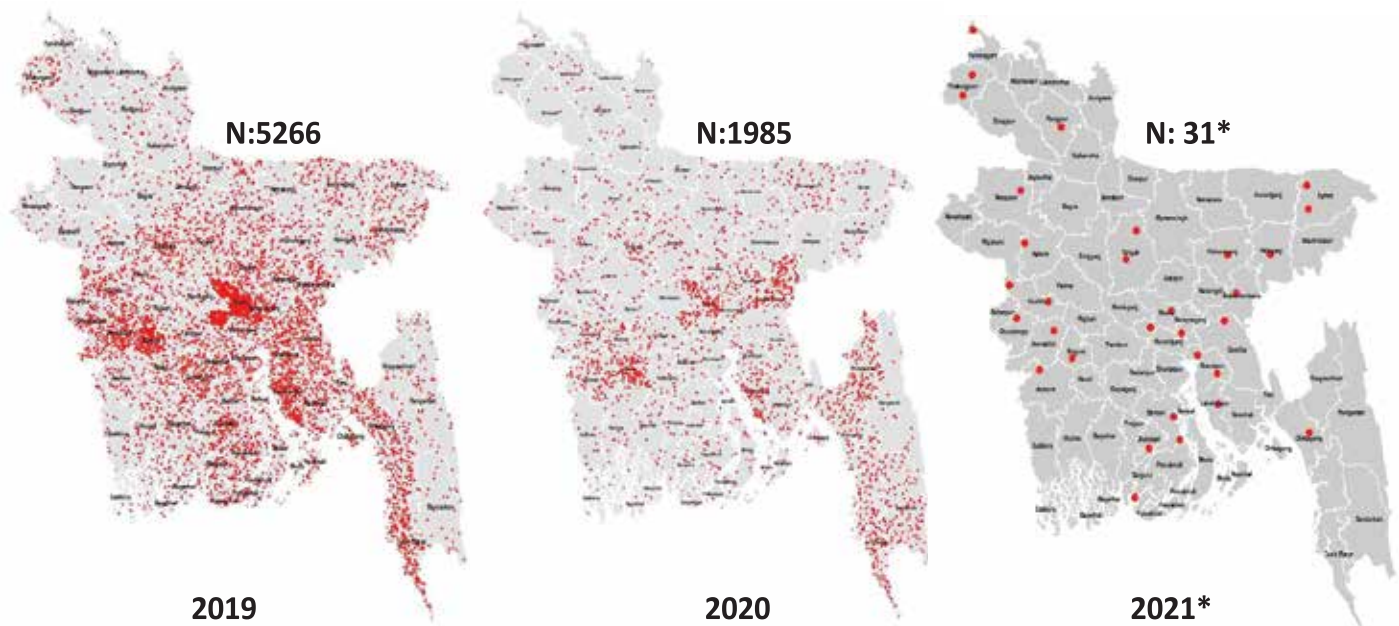
1 dot= 1 case



Dots are randomly placed within district

Distribution of Confirmed Measles Cases 2019, 2020 & 2021*

1 dot= 1 case



Confirmed Measles=Lab confirmed+Epi-linked cases

Dots are randomly placed within district

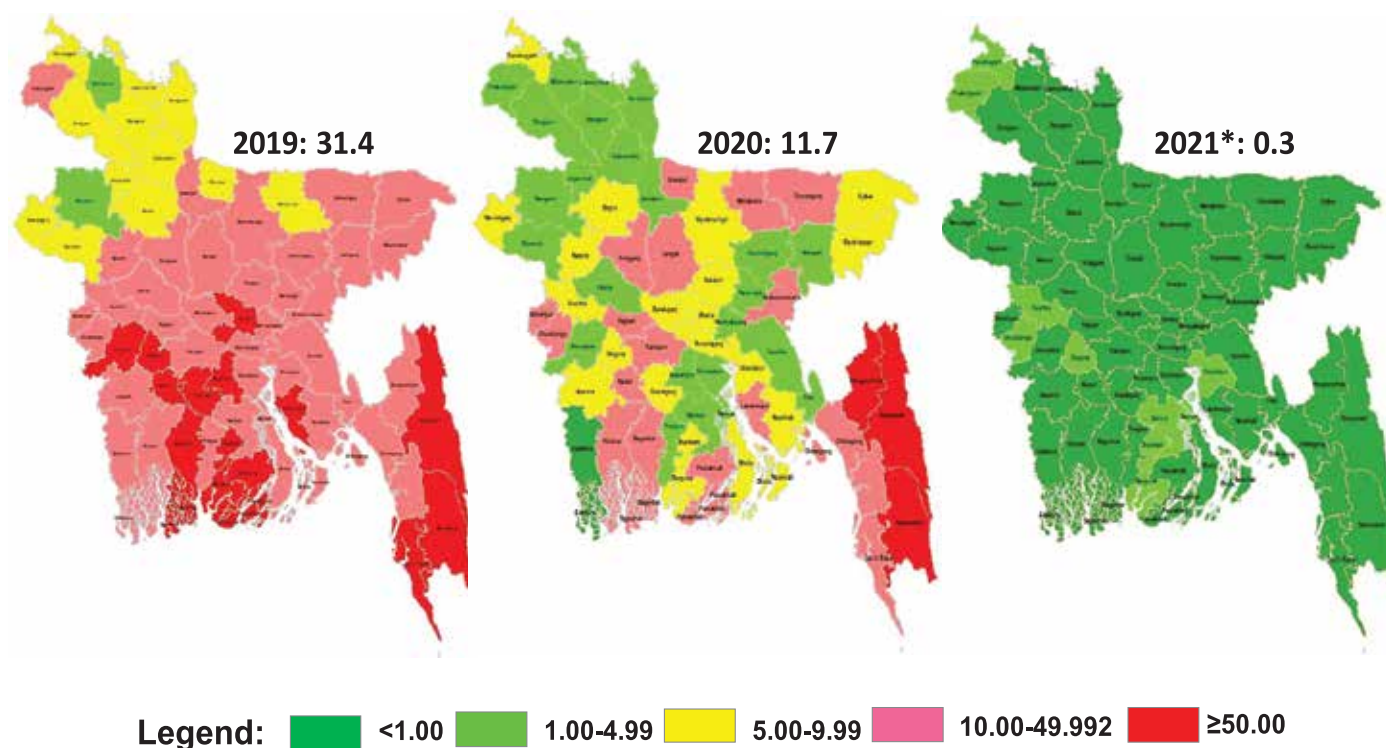
* Progress affected due to COVID-19

*Data as of 30 June 2021

MR bulletin, July 21

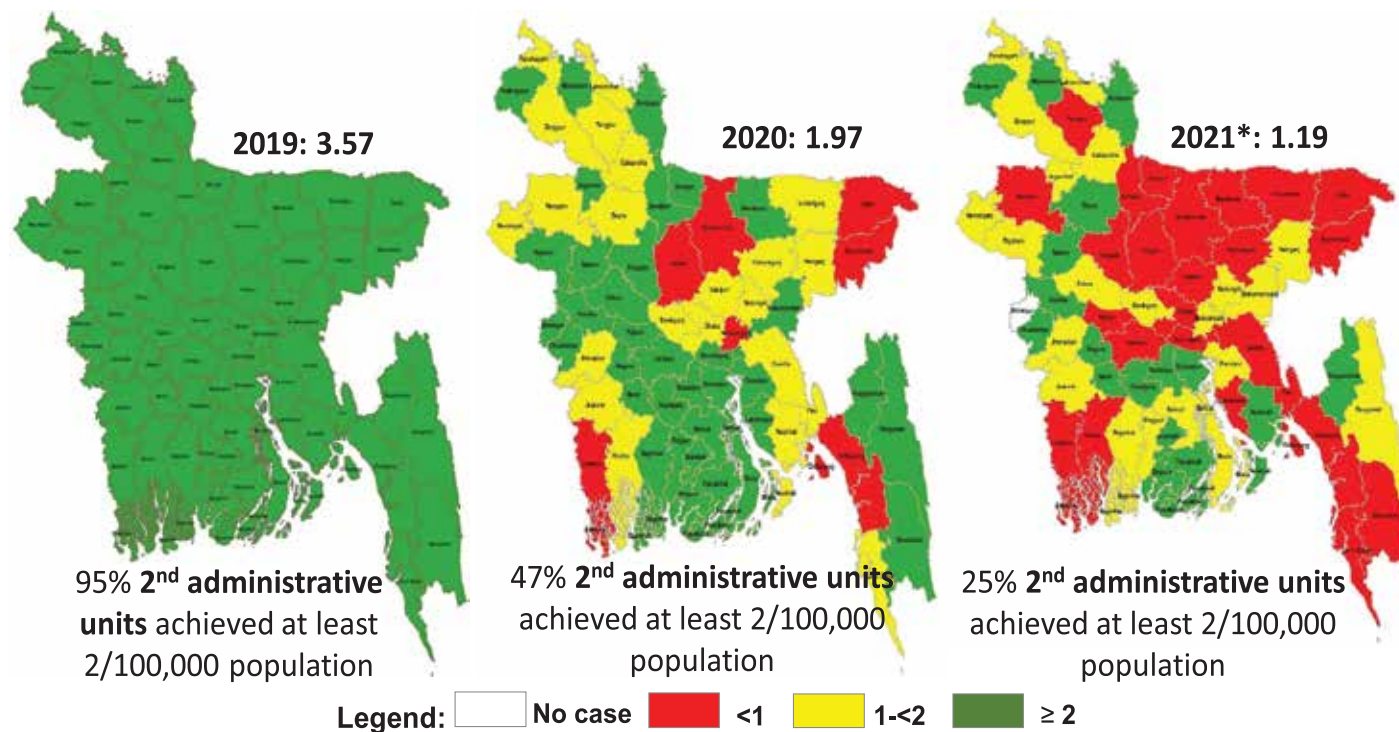
Data source: National EPI Surveillance

Measles Incidence per million population 2019-2021*



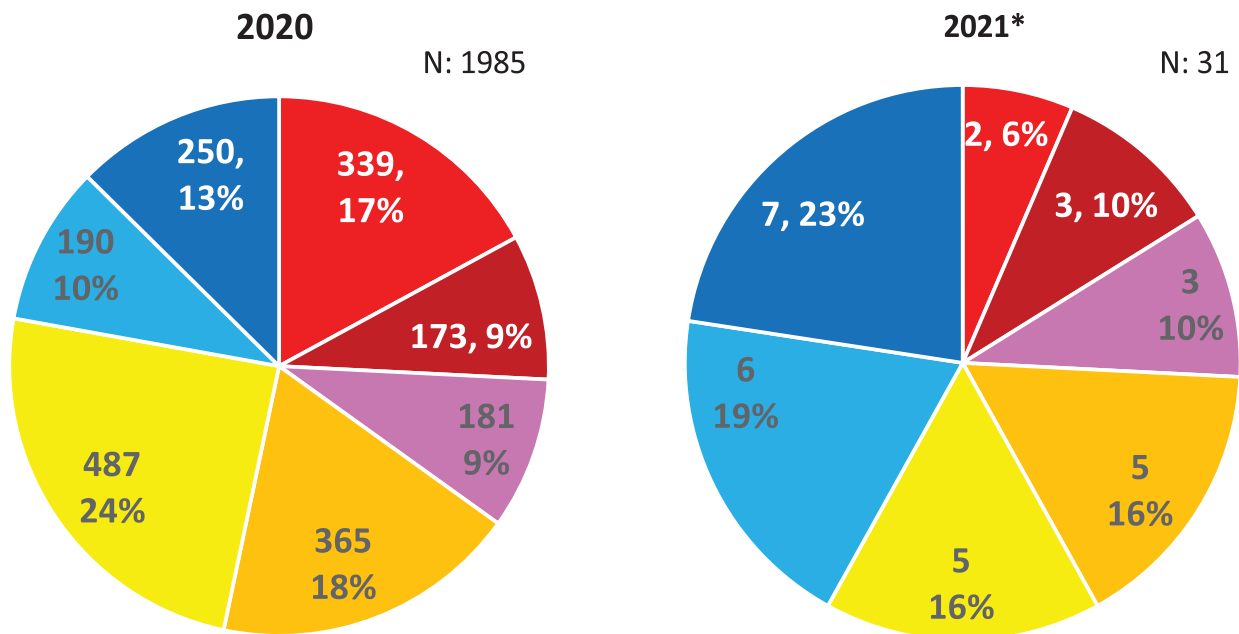
Non-measles non-rubella discarded rate per 100,000 population

National target: At least 2/100,000 population in > 80% of 2nd administrative units



*2nd administrative units (64 districts & 12 CC)

Age Distribution of Confirmed Measles Cases

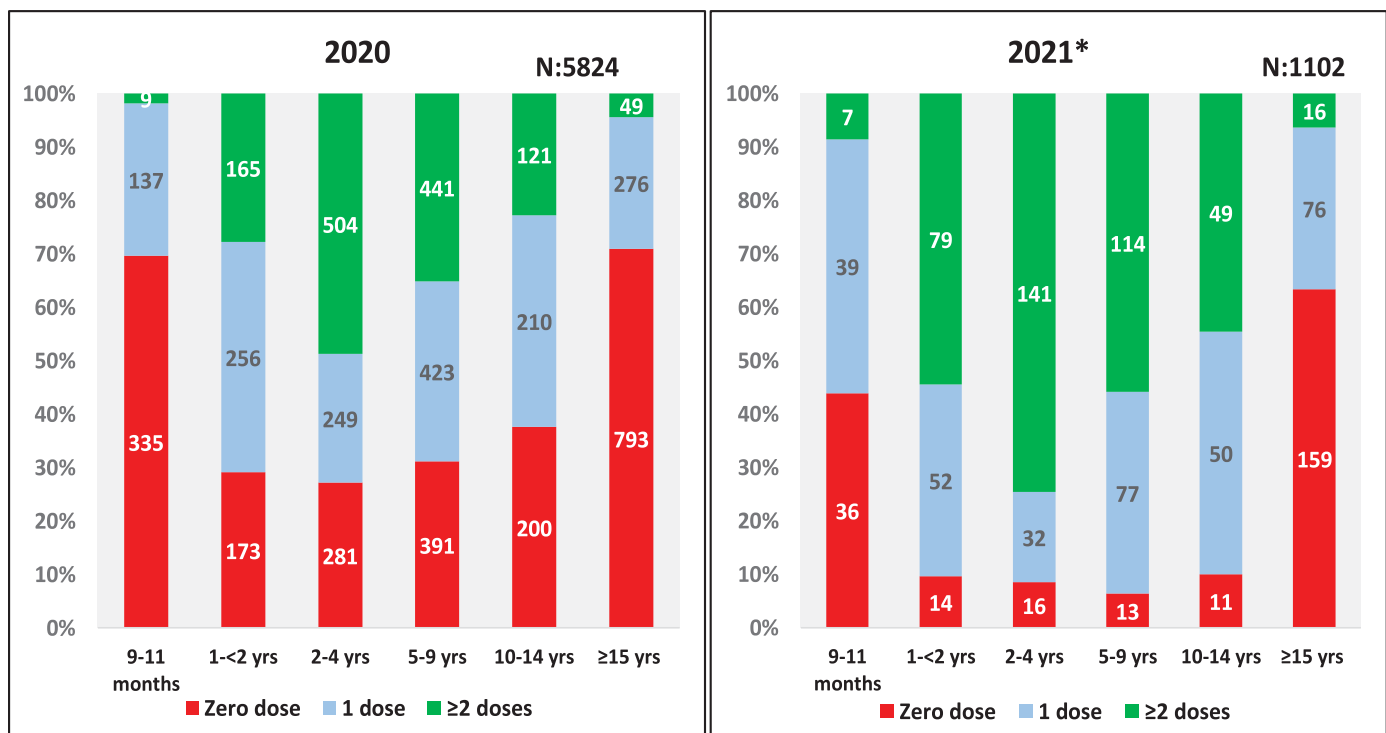


Confirmed Measles=Lab confirmed+Epi-linked cases

~ One third children affected are under 2 years of age

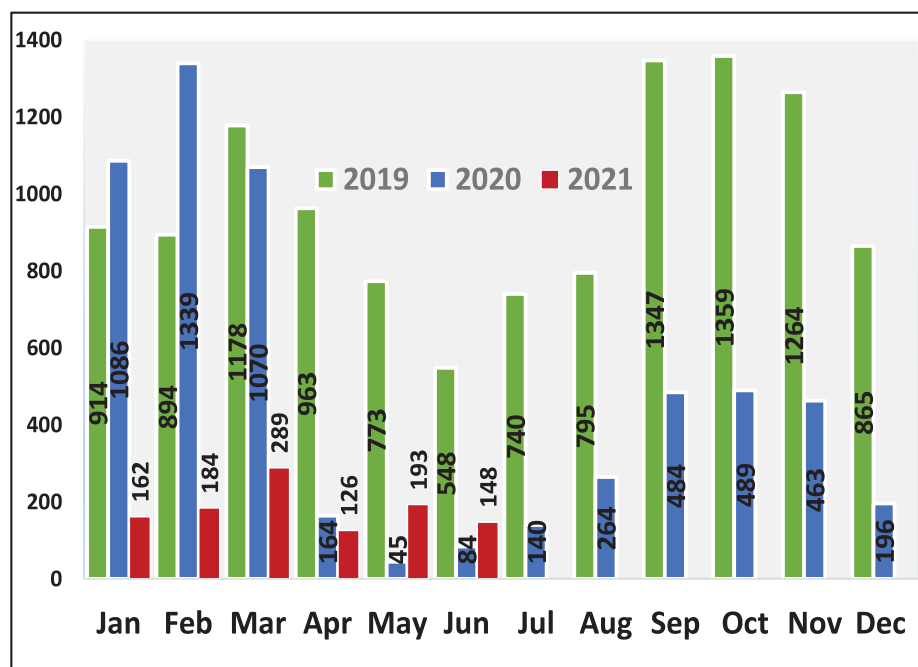
■ <9 months ■ 9-11 months ■ 1-2 yrs ■ 2-4 yrs ■ 5-9 yrs ■ 10-14 yrs ■ ≥15 yrs

Vaccination Status in Suspected Measles Cases



~ 60-65 cases under 2 years of age did not receive desired measles doses

Effect of Covid-19 on reporting of suspected measles case



Division	2020	2021*
BARISAL	447	123
CHITTAGONG	1513	187
DHAKA	1437	198
KHULNA	853	161
MYMENSINGH	208	33
RAJSHAHI	711	167
RANGPUR	446	179
SYLHET	209	54
Grand Total	5824	1102

Comparison of Suspected Measles Cases reported 2020 & 2021*



Legend: No Case <10 Cases 10-<50 Cases 50-<100 Cases ≥ 100 Cases

Distribution of Measles Outbreak 2019-2021*

1 dot= 1 outbreak

2019

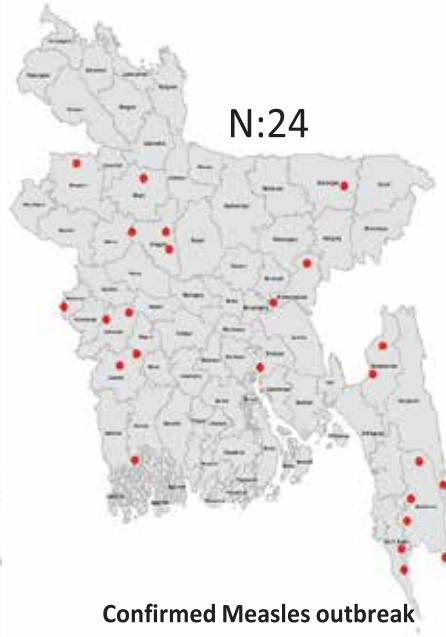
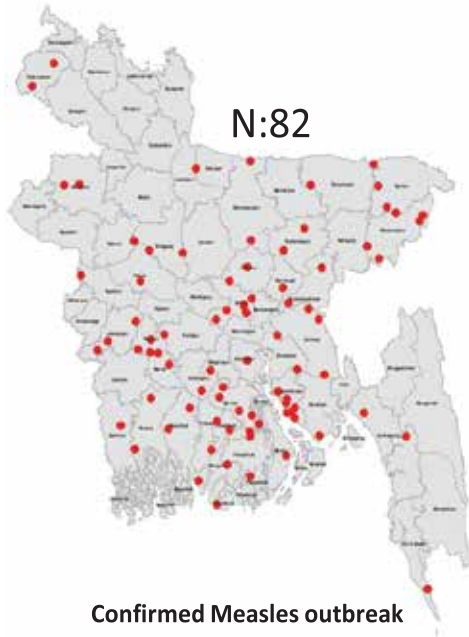
2020

2021*

N:82

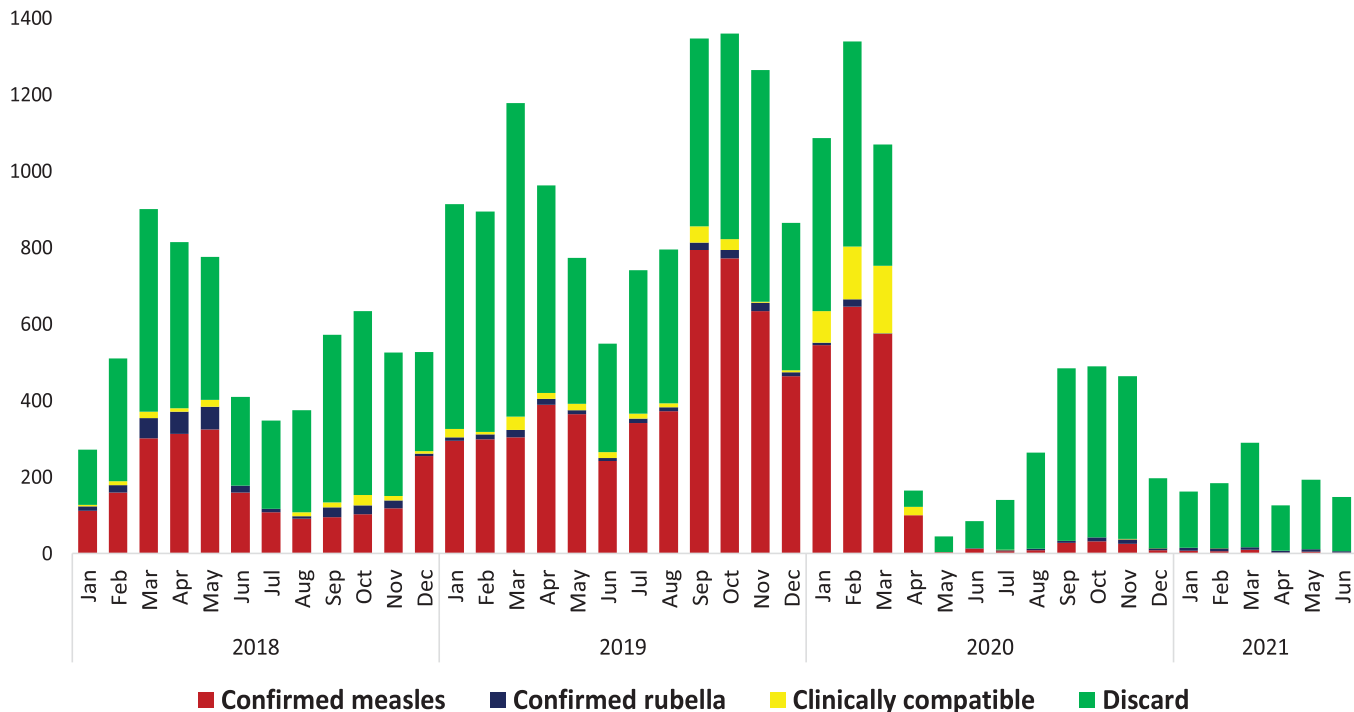
N:24

N:02



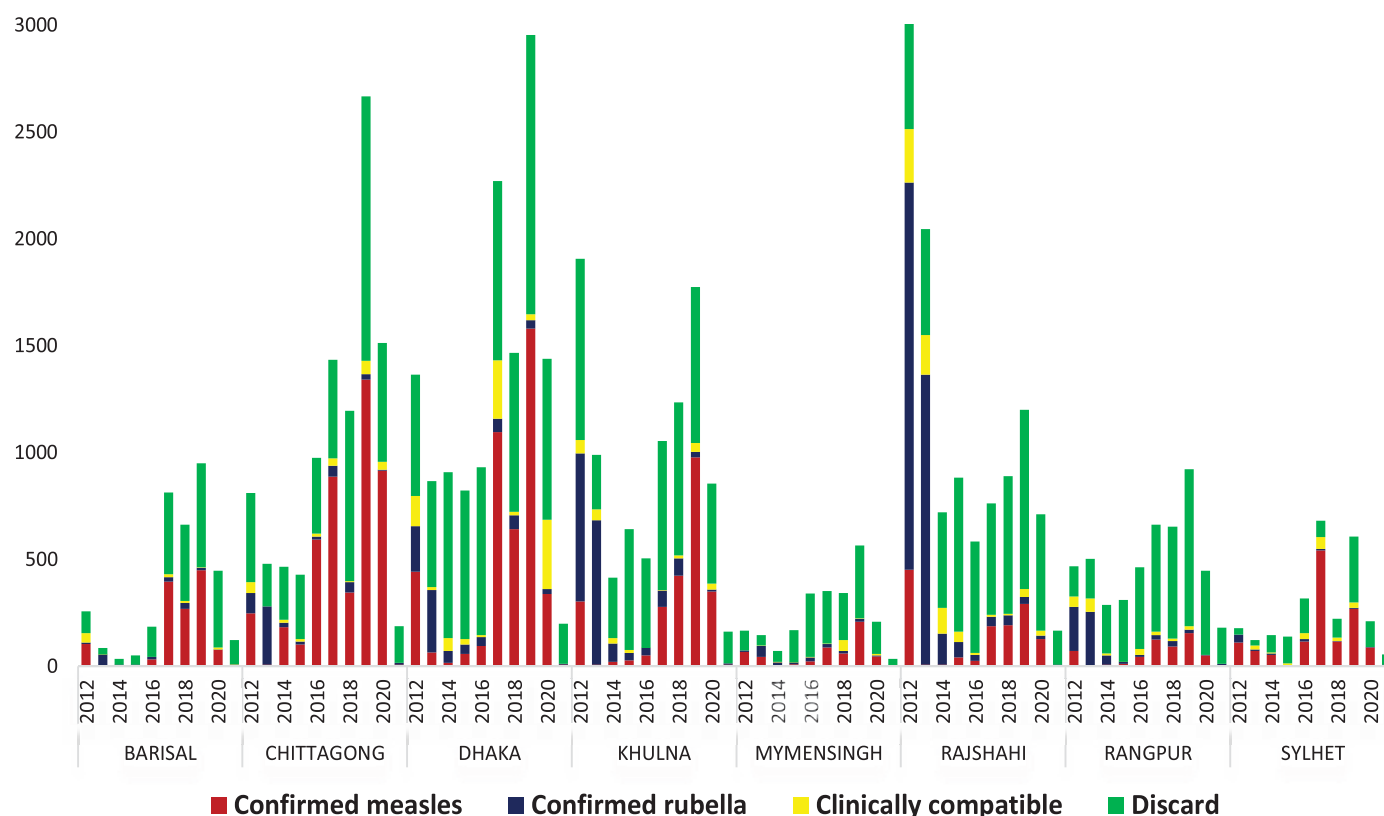
As of June-2021 total 14 suspected outbreak reported: Measles(2) Rubella(1), Discarded(8), Undefined(3)

Epicurve by case classification 2018-2021*



Dots are randomly placed within district

Measles case classification by Division from 2012-2021*



New Measles Elimination classification for SEARO Countries

1. Endemic

Existence of continuous transmission of measles and/or rubella virus, that persists for ≥ 12 months in any defined geographical area where measles had not been previously eliminated

2. Eliminated

Absence of endemic transmission for ≥ 12 months, but not verified by RVC

3. Verified

No endemic transmission for > 36 months in the presence of well performing surveillance system and verified by RVC

4. Re-established Endemic Transmission post verification:

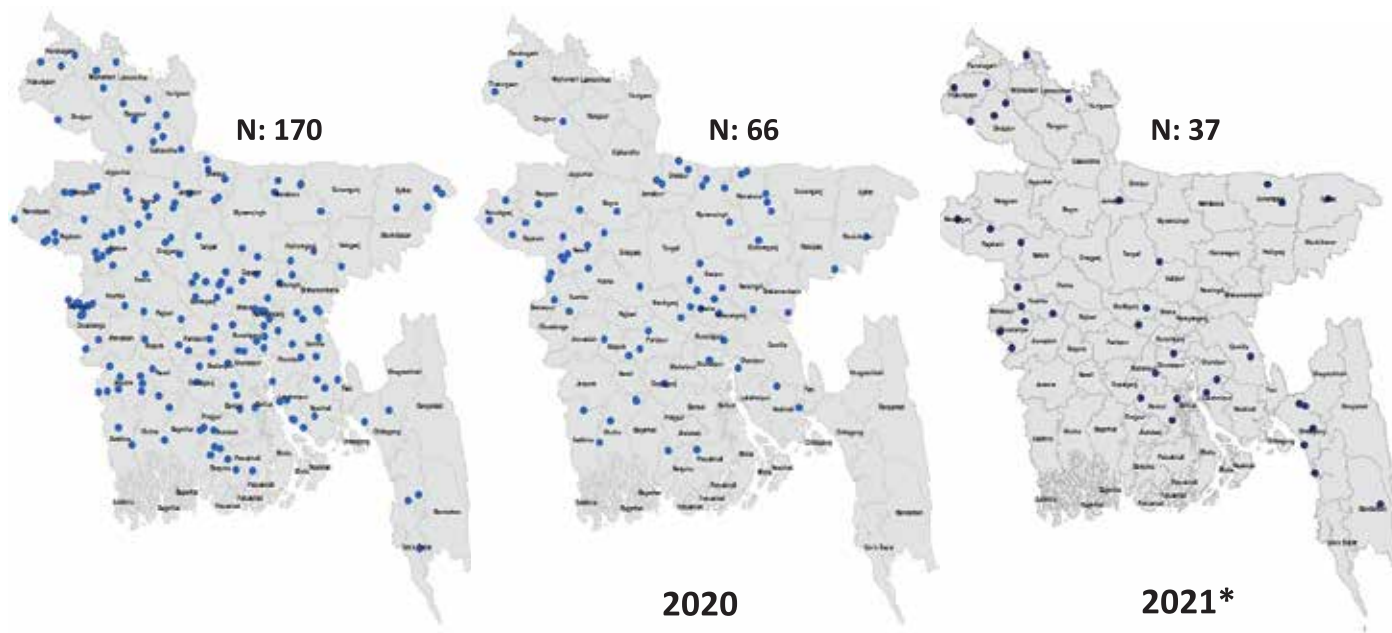
Presence of a chain of transmission of a virus strain that continues uninterrupted for ≥ 12 months in a defined geographical area where measles had previously been eliminated

The SEA-RVC members agreed that the **Rubella and CRS control status** has been maintained in the six countries but with the change of the Regional goal from control to elimination, the classification of “controlled” was withdrawn as it only denotes low level of transmission, but the virus is still endemic, and thus **Endemic** classification was adopted for these countries. However, this does not undermine the gains made by these countries to maintain the status of rubella control.

Rubella

Distribution of Confirmed Rubella Cases 2019, 2020 & 2021*

1 dot= 1 case



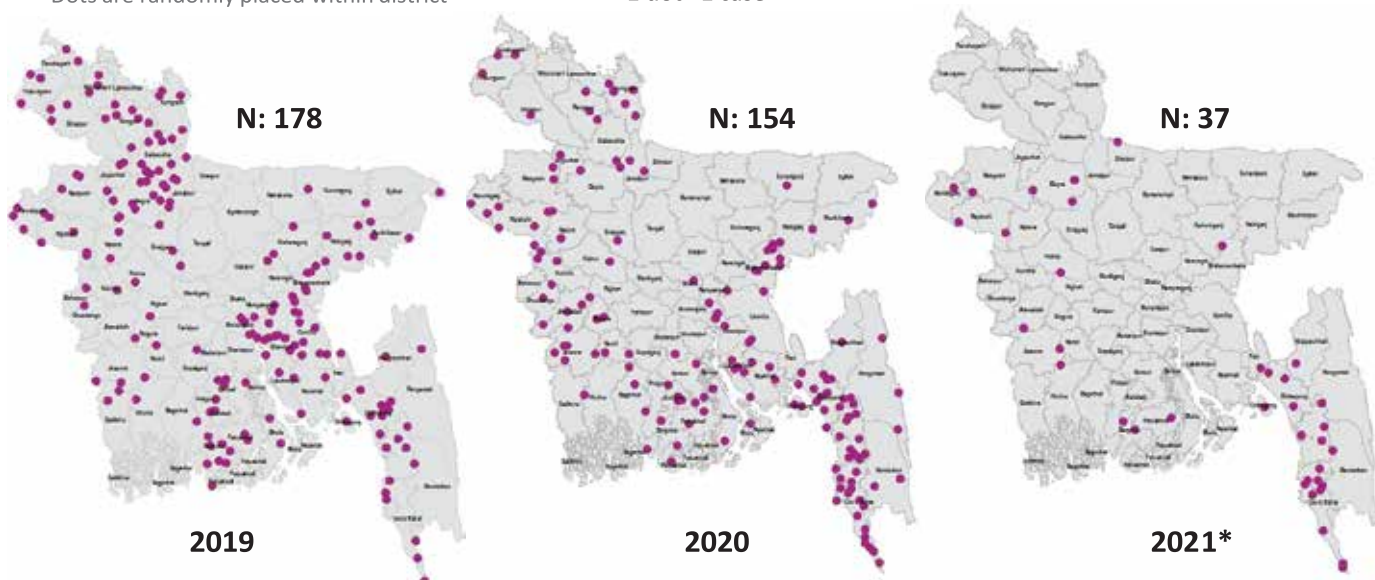
Dots are randomly placed within district

Congenital Rubella Syndrome (CRS)

Distribution of reported CRS Cases 2019, 2020 & 2021* (Suspected+Positive)

Dots are randomly placed within district

1 dot= 1 case



Year	Reported	Lab confirmed CRS	CRS rate /10,000 birth target ≥ 1 per 10 000livebirths
2017	201	11	0.61
2018	221	6	0.66
2019	178	3	0.52
2020	154	4	0.45
2021*	37	1	0.11

*Data as of 30 June 2021

MR bulletin, July 21

Data source: National EPI Surveillance

Summary of outbreaks 2020 & 2021*

Division	District	2020								2021*							
		# Investigated outbreaks	# Measles outbreaks	# Rubella outbreak	# Mixed outbreaks	# Undefined outbreaks	# Discarded outbreaks	# Outbreaks pendings for classification	#Deaths in outbreaks	# Investigated outbreaks	# Measles outbreaks	# Rubella outbreak	# Mixed outbreaks	# Undefined outbreaks	# Discarded outbreaks	# Outbreaks pendings for classification	#Deaths in outbreaks
Barisal	Barguna	1	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0
	Barisal	1	0	0	0	0	1	0	0	1	1	0	0	0	0	0	0
	Barisal CC	1	0	0	0	1	0	0	0	1	0	1	0	0	0	0	0
	Bhola	1	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0
	Jhalakati	1	0	0	0	0	1	0	0	1	0	0	0	1	0	0	0
	Patuakhali	2	0	0	0	0	2	0	0	0	0	0	0	0	0	0	0
	Perojpur	1	0	0	0	0	1	0	0	1	0	0	0	0	1	0	0
Barisal		8	0	0	0	2	6	0	0	4	1	1	0	1	1	0	0
Chittagong	Bandarban	6	6	0	0	0	0	0	1	0	0	0	0	0	0	0	0
	Brahmanbaria	4	2	0	1	0	1	0	0	1	0	0	0	1	0	0	0
	Chandpur	2	1	0	0	0	1	0	0	2	1	0	0	0	1	0	0
	Chittagong	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	Chittagong CC	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	Comilla	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	Comilla CC	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	Cox's Bazar	1	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0
	Feni	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	Khagrachari	2	2	0	0	0	0	0	1	0	0	0	0	0	0	0	0
	Lakshmipur	1	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0
	Noakhali	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	Rangamati	1	1	0	0	0	0	0	8	0	0	0	0	0	0	0	0
Chittagong		17	12	0	1	0	4	0	10	3	1	0	0	1	1	0	0
Dhaka	Dhaka	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0
	DNCC	1	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0
	DSCC	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	Faridpur	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	Gazipur	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	Gazipur CC	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	Gopalganj	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	Kishoreganj	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	Madaripur	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	Munshiganj	1	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0
	Narayanganj	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	Narayanganj CC	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	Narsingdi	1	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0
	Netrokona	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	Rajbari	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	Shariatpur	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	Tangail	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Dhaka		3	0	0	1	2	0	0	1	0	0	0	0	0	0	0	0
Khulna	Bagerhat	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	Chuadanga	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	Jessore	2	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	Jhenaidah	3	2	0	0	1	0	0	0	0	0	0	0	0	0	0	0
	Khulna	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	Khulna CC	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	Kushtia	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	Magura	1	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0
	Meherpur	1	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0
	Narail	2	1	0	1	0	0	0	0	0	0	0	0	0	0	0	0
	Satkhira	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Khulna		10	6	0	1	3	0	0	0	0	0	0	0	0	0	0	0

*Data as of 30 June 2021

MR bulletin, July 21

Data source: National EPI Surveillance

Summary of outbreaks 2020 & 2021*

Division	District	2020								2021*							
		# Investigated outbreaks	# Measles outbreaks	# Rubella outbreak	# Mixed outbreaks	# Undefined outbreaks	# Discarded outbreaks	# Outbreaks pendings for classification	#Deaths in outbreaks	# Investigated outbreaks	# Measles outbreaks	# Rubella outbreak	# Mixed outbreaks	# Undefined outbreaks	# Discarded outbreaks	# Outbreaks pendings for classification	#Deaths in outbreaks
Mymensingh	Jamalpur	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	Mymensingh	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	Mymensingh CC	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	Netrokona	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	Sherpur	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Mymensingh		0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Rajshahi	Bogra	2	1	0	1	0	0	0	0	2	0	0	0	0	2	0	0
	Joypurhat	2	0	0	0	0	2	0	0	0	0	0	0	0	0	0	0
	Naogaon	4	1	0	0	0	3	0	0	1	0	0	0	1	0	0	0
	Natore	0	0	0	0	0	0	0	0	1	0	0	0	0	1	0	0
	Nowabganj	2	0	0	0	0	2	0	0	1	0	0	0	0	1	0	0
	Pabna	5	0	0	0	1	4	0	0	1	0	0	0	0	1	0	0
	Rajshahi	1	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0
	Rajshahi CC	5	0	0	0	0	5	0	0	1	0	0	0	0	1	0	0
	Sirajganj	8	3	0	0	0	5	0	0	0	0	0	0	0	0	0	0
Rajshahi		29	5	0	1	1	22	0	0	7	0	0	0	1	6	0	0
Rangpur	Dinajpur	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	Gaibandha	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	Kurigram	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	Lalmonirhat	1	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0
	Nilphamari	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	Panchagarh	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	Rangpur	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	Rangpur CC	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	Thakurgaon	1	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0
Rangpur		2	0	0	0	1	1	0	0	0	0	0	0	0	0	0	0
Sylhet	Habiganj	1	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0
	Moulvi Bazar	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	Sunamganj	1	1	0	0	0	0	1	0	0	0	0	0	0	0	0	0
	Sylhet	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	Sylhet CC	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Sylhet		2	1	0	0	0	1	0	0	0	0	0	0	0	0	0	0
Grand Total		71	24	0	4	9	34	0	11	14	2	1	0	3	8	0	0

Specimens for measles surveillance in Bangladesh

For Serology: 3-5 ml blood for older children and adults and 1 ml for infants and younger children should be collected within 2-28 days of onset of rash

- Whole blood can be stored at 4–8 °C for up to 24 hours before the serum is separated.
- Serum should be stored at 4–8°C until shipment takes place, but not more than a maximum of 5 days. When kept for longer periods (in unavoidable situation), serum samples must be frozen at a temperature of –20°C.

For Virology: Throat swabs should be collected within 5 days of onset of rash

- Throat swab specimens should be refrigerated and shipped to the laboratory with ice packs (4 to 8°C) to arrive at the testing laboratory within 48 hours.

Summary of Cases 2020

Division	District	Measles Cases				Clinically compatible	Rubella Cases			Discarded Cases	Cases pending for classification	Incidence		Discarded Rate
		Total cases	Lab confirmed	Epi-linked	Total		Lab confirmed	Epi-linked	Total	Both Negative (Discarded)		Incidence per 1 million population		Non-measles non-rubella discarded rate per 100,000 pop
												Measles	Rubella	District status
BARISAL	BARGUNA	73	8	0	8	1	0	0	0	64	0	7.58	0.00	6.06
	BARISAL	67	7	0	7	2	0	0	0	58	0	2.97	0.00	2.46
	BARISAL CC.	20	2	0	2	2	0	0	0	16	0	5.15	0.00	4.12
	BHOLA	88	18	0	18	1	0	0	0	69	0	8.57	0.00	3.28
	JHALAKATI	34	6	0	6	1	0	0	0	27	0	7.43	0.00	3.35
	PATUAKHALI	113	29	0	29	1	1	0	1	82	0	15.97	0.55	4.52
	PEROJPUR	52	6	0	6	1	1	0	1	44	0	4.56	0.76	3.34
BARISAL	447	76	0	76	9	2	0	2	360	0	7.72	0.20	3.66	
CHITTAGONG	BANDARBAN	113	36	55	91	0	0	0	0	22	0	198.20	0.00	4.79
	BRAHMANBARIA	207	25	98	123	2	2	0	2	80	0	36.62	0.60	2.38
	CHANDPUR	108	23	0	23	11	1	0	1	73	0	8.05	0.35	2.56
	CHITTAGONG	77	43	0	43	0	0	0	0	34	0	7.23	0.00	0.57
	CHITTAGONG CC.	90	56	0	56	0	0	0	0	34	0	18.33	0.00	1.11
	COMILLA	129	24	0	24	8	0	0	0	97	0	4.01	0.00	1.62
	COMILLA CC.	6	0	0	0	0	0	0	0	6	0	0.00	0.00	1.55
	COX'S BAZAR	133	87	0	87	1	0	0	0	45	0	32.13	0.00	1.66
	FENI	25	6	0	6	0	0	0	0	19	0	3.53	0.00	1.12
	KHAGRACHARI	142	15	92	107	0	0	0	0	35	0	147.41	0.00	4.82
	LAKSHMIPUR	120	66	0	66	6	0	0	0	48	0	32.28	0.00	2.35
	NOAKHALI	93	34	0	34	7	2	0	2	50	0	9.25	0.54	1.36
RANGAMATI	270	7	248	255	0	0	0	0	15	0	361.89	0.00	2.13	
CHITTAGONG	1513	422	493	915	35	5	0	5	558	0	27.23	0.15	1.66	
DHAKA	DHAKA	96	25	0	25	32	0	0	0	39	0	4.22	0.00	0.66
	DHAKA NORTH CC.	222	45	0	45	104	2	0	2	71	0	9.62	0.43	1.52
	DHAKA SOUTH CC.	186	47	0	47	81	2	0	2	56	0	12.92	0.55	1.54
	FARIDPUR	94	36	0	36	3	2	0	2	53	0	15.92	0.88	2.34
	GAZIPUR	71	21	0	21	12	1	0	1	37	0	9.80	0.47	1.73
	GAZIPUR CC.	43	12	0	12	11	2	0	2	18	0	6.38	1.06	0.96
	GOPALGANJ	67	13	0	13	4	1	0	1	49	0	9.38	0.72	3.53
	KISHOREGANJ	84	13	0	13	7	1	0	1	63	0	3.78	0.29	1.83
	MADARIPUR	72	10	0	10	7	1	0	1	54	0	7.25	0.73	3.92
	MANIKGANJ	48	5	0	5	2	0	0	0	41	0	3.04	0.00	2.49
	MUNSHIGANJ	54	10	0	10	14	1	0	1	29	0	5.85	0.59	1.70
	NARAYANGANJ	78	12	0	12	28	1	0	1	37	0	4.53	0.38	1.40
	NARAYANGANJ CC.	7	3	0	3	0	0	0	0	4	0	3.58	0.00	0.48
	NARSINGDHI	109	30	11	41	11	3	0	3	54	0	15.59	1.14	2.05
	RAJBARI	38	5	0	5	1	1	0	1	31	0	4.03	0.81	2.50
	SARIATPUR	55	7	0	7	3	0	0	0	45	0	5.12	0.00	3.29
	TANGAIL	113	32	0	32	4	6	0	6	71	0	7.51	1.41	1.67
DHAKA	1437	326	11	337	324	24	0	24	752	0	7.82	0.56	1.75	
KHULNA	BAGERHAT	80	24	0	24	0	0	0	0	56	0	13.75	0.00	3.21
	CHUADANGA	44	6	0	6	0	1	0	1	37	0	4.49	0.75	2.77
	JESSORE	143	50	16	66	3	3	0	3	71	0	20.19	0.92	2.17
	JHENAIDAH	92	38	23	61	1	0	0	0	30	0	29.13	0.00	1.43
	KHULNA	59	12	15	27	0	0	0	0	32	0	13.80	0.00	1.64
	KHULNA CC.	11	6	0	6	0	0	0	0	5	0	7.65	0.00	0.64
	KUSHTIA	99	18	0	18	2	1	0	1	78	0	7.82	0.43	3.39
	MAGURA	77	23	0	23	4	0	0	0	50	0	21.18	0.00	4.60
	MEHERPUR	47	7	0	7	16	0	0	0	24	0	9.03	0.00	3.10
	NARAIL	136	61	33	94	0	4	0	4	38	0	110.17	4.69	4.45
	SATKHIRA	65	17	0	17	0	1	0	1	47	0	7.24	0.43	2.00
KHULNA	853	262	87	349	26	10	0	10	468	0	18.82	0.54	2.52	

Summary of Cases 2020

Division	District	Measles Cases				Clinically compatible	Rubella Cases			Discarded Cases	Cases pending for classification	Incidence		Discarded Rate
		Total cases	Lab confirmed	Epi-linked	Total		Lab confirmed	Epi-linked	Total			Both Negative (Discarded)	Incidence per 1 million population	
										Measles			Rubella	District status
MYMENSINGH	JAMALPUR	54	6	0	6	1	2	0	2	45	0	2.21	0.74	1.66
	MYMENSINGH	80	23	0	23	7	1	0	1	49	0	4.01	0.17	0.85
	MYMENSINGH CC.	4	4	0	4	0	0	0	0	0	0	13.11	0.00	0.00
	NETROKONA	54	11	0	11	0	0	0	0	43	0	4.17	0.00	1.63
	SHERPUR	16	2	0	2	0	1	0	1	13	0	1.25	0.62	0.81
MYMENSINGH		208	46	0	46	8	4	0	4	150	0	3.54	0.31	1.15
RAJSHAHI	BOGRA	90	10	16	26	0	1	0	1	63	0	6.47	0.25	1.57
	JOYPURHAT	37	3	0	3	0	0	0	0	34	0	2.78	0.00	3.15
	NATORE	84	16	0	16	0	6	0	6	62	0	7.93	2.97	3.07
	NOAGOAN	65	7	2	9	1	2	0	2	53	0	2.93	0.65	1.72
	NOWABGANJ	49	10	0	10	1	2	0	2	36	0	5.13	1.03	1.85
	PABNA	137	9	0	9	20	2	0	2	106	0	3.02	0.67	3.55
	RAJSHAHI	53	6	0	6	0	3	0	3	44	0	2.37	1.18	1.73
	RAJSHAHI CC.	70	6	0	6	0	0	0	0	64	0	11.28	0.00	12.04
SIRAJGANJ	126	15	26	41	1	0	0	0	84	0	11.20	0.00	2.29	
RAJSHAHI		711	82	44	126	23	16	0	16	546	0	5.77	0.73	2.50
RANGPUR	DINAJPUR	54	6	0	6	0	1	0	1	47	0	1.70	0.28	1.33
	GAIBANDHA	54	6	0	6	0	0	0	0	48	0	2.13	0.00	1.71
	KURIGRAM	95	7	0	7	0	0	0	0	88	0	2.86	0.00	3.60
	LALMONIRHAT	25	3	0	3	0	0	0	0	22	0	2.02	0.00	1.48
	NILPHAMARI	65	9	0	9	0	0	0	0	56	0	4.15	0.00	2.58
	PANCHAGHARH	25	7	0	7	0	0	0	0	18	0	5.99	0.00	1.54
	RANGPUR	42	8	0	8	0	0	0	0	34	0	2.62	0.00	1.11
	RANGPUR CC.	3	1	0	1	0	0	0	0	2	0	2.87	0.00	0.57
THAKURGOAN	83	2	0	2	0	2	0	2	79	0	1.22	1.22	4.81	
RANGPUR		446	49	0	49	0	3	0	3	394	0	2.63	0.16	2.11
SYLHET	HABIGANJ	47	3	0	3	0	0	0	0	44	0	1.21	0.00	1.78
	MOULVI BAZAR	37	16	0	16	0	2	0	2	19	0	7.05	0.88	0.84
	SUNAMGANJ	80	44	3	47	0	0	0	0	33	0	16.11	0.00	1.13
	SYLHET	43	19	0	19	0	0	0	0	24	0	5.44	0.00	0.69
	SYLHET CC.	2	2	0	2	0	0	0	0	0	0	3.53	0.00	0.00
SYLHET		209	84	3	87	0	2	0	2	120	0	7.43	0.17	1.02
Grand Total		5824	1347	638	1985	425	66	0	66	3348	0	11.7	0.39	1.97

Importance of Sample collection in Measles surveillance

Serum: As the country goal is to eliminate Measles, Rubella & CRS, every case should be classified** based on antibody testing (IgM ELISA).

Throat swab: Throat swab is the preferable sample for virology testing (genotyping). For elimination setup, along with antibody testing, data on viral genotypes are critical for identifying the source of cases, whether they are indigenous or imported, and the place of origin if imported.

Summary of Cases 2021*

Division	District		Measles Cases				Clinically compatible	Rubella Cases			Discarded Cases	Cases pending for classification	Incidence		Discarded Rate
			Total cases	Lab confirmed	Epi-linked	Total		Lab confirmed	Epi-linked	Total			Both Negative (Discarded)	Incidence per 1 million population	
											Measles			Rubella	District status
BARISAL	BARGUNA	1071070	26	1	0	1	0	0	0	0	25	0	1.87	0.00	4.67
	BARISAL	2394642	21	2	0	2	0	0	0	0	19	0	1.67	0.00	1.59
	BARISAL CC.	393835	10	0	0	0	0	3	0	3	7	0	0.00	15.23	3.55
	BHOLA	2131623	17	0	0	0	0	0	0	0	17	0	0.00	0.00	1.60
	JHALAKATI	818999	15	1	0	1	1	0	0	0	13	0	2.44	0.00	3.17
	PATUAKHALI	1842566	22	0	0	0	0	0	0	0	22	0	0.00	0.00	2.39
	PEROJPUR	1335576	12	0	0	0	0	0	0	0	12	0	0.00	0.00	1.80
BARISAL	9988311	123	4	0	4	1	3	0	3	115	0	0.80	0.60	2.30	
CHITTAGONG	BANDARBAN	465886	2	0	0	0	0	1	0	1	1	0	0.00	4.29	0.43
	BRAHMANBARIA	3407749	32	1	0	1	0	0	0	0	31	0	0.59	0.00	1.82
	CHANDPUR	2898499	23	2	0	2	0	0	0	0	21	0	1.38	0.00	1.45
	CHITTAGONG	6036513	15	1	0	1	0	2	0	2	12	0	0.33	0.66	0.40
	CHITTAGONG CC.	3100833	13	0	0	0	0	3	0	3	10	0	0.00	1.93	0.64
	COMILLA	6071570	19	1	0	1	0	1	0	1	17	0	0.33	0.33	0.56
	COMILLA CC.	391566	0	0	0	0	0	0	0	0	0	0	0.00	0.00	0.00
	COX'S BAZAR	2747304	2	0	0	0	0	0	0	0	2	0	0.00	0.00	0.15
	FENI	1724416	4	0	0	0	0	0	0	0	4	0	0.00	0.00	0.46
	KHAGRACHARI	736517	14	0	0	0	0	0	0	0	14	0	0.00	0.00	3.80
	LAKSHMIPUR	2074509	13	1	0	1	0	2	0	2	10	0	0.96	1.93	0.96
	NOAKHALI	3728771	46	0	0	0	0	0	0	0	46	0	0.00	0.00	2.47
RANGAMATI	714997	4	0	0	0	0	0	0	0	4	0	0.00	0.00	1.12	
CHITTAGONG	34099128	187	6	0	6	0	9	0	9	172	0	0.35	0.53	1.01	
DHAKA	DHAKA	6011585	12	2	0	2	0	1	0	1	9	0	0.67	0.33	0.30
	DHAKA NORTH CC.	4746380	16	0	0	0	0	0	0	0	16	0	0.00	0.00	0.67
	DHAKA SOUTH CC.	3691208	5	0	0	0	0	1	0	1	4	0	0.00	0.54	0.22
	FARIDPUR	2294991	2	0	0	0	0	0	0	0	2	0	0.00	0.00	0.17
	GAZIPUR	2173689	18	0	0	0	0	0	0	0	18	0	0.00	0.00	1.66
	GAZIPUR CC.	1909988	2	0	0	0	0	0	0	0	2	0	0.00	0.00	0.21
	GOPALGANJ	1406548	20	0	0	0	0	0	0	0	20	0	0.00	0.00	2.84
	KISHOREGANJ	3493418	17	1	0	1	0	0	0	0	16	0	0.57	0.00	0.92
	MADARIPUR	1398794	17	0	0	0	0	1	0	1	16	0	0.00	1.43	2.29
	MANIKGANJ	1671024	10	0	0	0	0	0	0	0	10	0	0.00	0.00	1.20
	MUNSHIGANJ	1734360	5	0	0	0	0	0	0	0	5	0	0.00	0.00	0.58
	NARAYANGANJ	2685934	22	1	0	1	0	0	0	0	21	0	0.74	0.00	1.56
	NARAYANGANJ CC.	851045	2	0	0	0	0	0	0	0	2	0	0.00	0.00	0.47
	NARSINGDHI	2669268	18	0	0	0	0	0	0	0	18	0	0.00	0.00	1.35
	RAJBARI	1259420	4	0	0	0	0	0	0	0	4	0	0.00	0.00	0.64
	SARIATPUR	1386643	15	0	0	0	0	1	0	1	14	0	0.00	1.44	2.02
	TANGAIL	4325022	12	2	0	2	0	0	0	0	10	0	0.92	0.00	0.46
DHAKA	43709317	197	6	0	6	0	4	0	4	187	0	0.27	0.18	0.86	
KHULNA	BAGERHAT	1770867	13	0	0	0	0	0	0	0	13	0	0.00	0.00	1.47
	CHUADANGA	1354481	21	1	0	1	0	3	0	3	17	0	1.48	4.43	2.51
	JESSORE	3316630	23	1	0	1	0	0	0	0	22	0	0.60	0.00	1.33
	JHENAI DAH	2125035	13	1	0	1	0	0	0	0	12	0	0.94	0.00	1.13
	KHULNA	1985727	2	0	0	0	0	0	0	0	2	0	0.00	0.00	0.20
	KHULNA CC.	795812	5	0	0	0	0	0	0	0	5	0	0.00	0.00	1.26
	KUSHTIA	2335624	31	2	0	2	0	3	0	3	26	0	1.71	2.57	2.23
	MAGURA	1101828	33	1	0	1	0	0	0	0	32	0	1.82	0.00	5.81
	MEHERPUR	786275	0	0	0	0	0	0	0	0	0	0	0.00	0.00	0.00
	NARAIL	865786	9	0	0	0	0	0	0	0	9	0	0.00	0.00	2.08
SATKHIRA	2382557	11	0	0	0	0	0	0	0	11	0	0.00	0.00	0.92	
KHULNA	18820622	161	6	0	6	0	6	0	6	149	0	0.64	0.64	1.58	

Summary of Cases 2021*

Division	District		Measles Cases				Clinically compatible	Rubella Cases			Discarded Cases	Cases pending for classification	Incidence		Discarded Rate
			Total cases	Lab confirmed	Epi-linked	Total		Lab confirmed	Epi-linked	Total			Both Negative (Discarded)	Incidence per 1 million population	
											Measles			Rubella	District status
MYMENSINGH	JAMALPUR	2750524	7	0	0	0	0	1	0	1	6	0	0.00	0.73	0.44
	MYMENSINGH	5821228	12	0	0	0	0	1	0	1	11	0	0.00	0.34	0.38
	MYMENSINGH CC.	309571	1	0	0	0	0	0	0	0	1	0	0.00	0.00	0.65
	NETROKONA	2674904	11	0	0	0	0	0	0	0	11	0	0.00	0.00	0.82
	SHERPUR	1629584	2	0	0	0	0	0	0	0	2	0	0.00	0.00	0.25
MYMENSINGH	13185810	33	0	0	0	0	2	0	2	31	0	0.00	0.30	0.47	
RAJSHAHI	BOGRA	4080032	47	0	0	0	0	0	0	0	47	0	0.00	0.00	2.30
	JOYPURHAT	1096248	8	0	0	0	0	0	0	0	8	0	0.00	0.00	1.46
	NATORE	2047498	33	1	0	1	0	0	0	0	32	0	0.98	0.00	3.13
	NOAGOAN	3119411	14	1	0	1	0	0	0	0	13	0	0.64	0.00	0.83
	NOWABGANJ	1976533	15	0	0	0	0	1	0	1	14	0	0.00	1.01	1.42
	PABNA	3027061	18	0	0	0	0	0	0	0	18	0	0.00	0.00	1.19
	RAJSHAHI	2573888	17	0	0	0	0	2	0	2	15	0	0.00	1.55	1.17
	RAJSHAHI CC.	539573	10	0	0	0	0	0	0	0	10	0	0.00	0.00	3.71
	SIRAJGANJ	3716061	5	0	0	0	0	0	0	0	5	0	0.00	0.00	0.27
RAJSHAHI	22176305	167	2	0	2	0	3	0	3	162	0	0.18	0.27	1.46	
RANGPUR	DINAJPUR	3587260	25	0	0	0	0	2	0	2	23	0	0.00	1.12	1.28
	GAIBANDHA	2854395	23	0	0	0	0	0	0	0	23	0	0.00	0.00	1.61
	KURIGRAM	2482509	25	0	0	0	0	0	0	0	25	0	0.00	0.00	2.01
	LALMONIRHAT	1506943	12	0	0	0	0	2	0	2	10	0	0.00	2.65	1.33
	NILPHAMARI	2200529	26	0	0	0	0	0	0	0	26	0	0.00	0.00	2.36
	PANCHAGHARH	1184878	11	1	0	1	0	0	0	0	10	0	1.69	0.00	1.69
	RANGPUR	3103412	18	1	0	1	0	0	0	0	17	0	0.64	0.00	1.10
	RANGPUR CC.	353030	0	0	0	0	0	0	0	0	0	0	0.00	0.00	0.00
	THAKURGOAN	1667635	39	2	0	2	0	3	0	3	34	0	2.40	3.60	4.08
RANGPUR	18940591	179	4	0	4	0	7	0	7	168	0	0.42	0.74	1.77	
SYLHET	HABIGANJ	2506177	15	1	0	1	0	0	0	0	14	0	0.80	0.00	1.12
	MOULVI BAZAR	2302301	4	0	0	0	0	0	0	0	4	0	0.00	0.00	0.35
	SUNAMGANJ	2960824	13	0	0	0	0	2	0	2	11	0	0.00	1.35	0.74
	SYLHET	3544338	20	2	0	2	0	1	0	1	17	0	1.13	0.56	0.96
	SYLHET CC.	575661	2	0	0	0	0	0	0	0	2	0	0.00	0.00	0.69
SYLHET	11889301	54	3	0	3	0	3	0	3	48	0	0.50	0.50	0.81	
Grand Total		172809384	1101	31	0	31	1	37	0	37	1032	0	0.36	0.43	1.19

Cases Investigation within 5 days of Rash onset vs Swab Sample Collection 2019-2021*

Division	2019		2020		2021*	
	Case investigation within 5 days of rash onset	SWAB sample collected	Case investigation within 5 days of rash onset	SWAB sample collected	Case investigation within 5 days of rash onset	SWAB sample collected
BARISAL	450	7	199	1	54	0
CHITTAGONG	1265	122	710	11	117	6
DHAKA	1684	52	984	2	149	0
KHULNA	833	207	384	48	82	2
MYMENSINGH	257	8	125	0	19	0
RAJSHAHI	453	41	243	14	86	4
RANGPUR	488	70	278	8	129	1
SYLHET	338	1	145	1	32	0
Total	5768	508	3068	85	668	13

Key Definitions

- **Suspected measles case (New Case Definition):** Any person in whom a clinician suspects measles infection or any person with **fever and generalized maculopapular** (i.e., non-vesicular) rash.
- **Laboratory-confirmed measles, rubella and CRS case:** A case that meets the case definition and has laboratory confirmation of measles/rubella infection and/or isolation of virus in a clinical sample by WHO accredited national laboratory.
- **Epidemiologically-linked confirmed measles, or rubella, case:** A case that meets the clinical case definition and is linked epidemiologically to a laboratory confirmed case or another epidemiologically confirmed case.
- **Clinically compatible measles:** Any person meeting the clinical case definition in whom a laboratory investigation is not conducted and does not have epidemiologic linkage to a laboratory confirmed case.
- **Clinically compatible rubella:** Not applicable.
- **Suspected CRS:** Any infant, less than 1 year of age that a health worker suspects of having CRS.
A health worker should suspect CRS when
 - ✓ an infant aged 0–11 months presents with heart disease and/or suspicion of hearing impairment and/or one or more of the following eye signs: white pupil (cataract), larger eyeball (congenital glaucoma) or pigmentary retinopathy.
 - ✓ an infant's mother has a history of suspected or confirmed rubella during pregnancy, even when the infant shows no signs of CRS.
- **Clinically compatible CRS case:** An infant in whom a qualified physician detects at least two of the findings listed in (a) or one in (a) and one in (b) and without any other obvious clinical cause that has not been adequately tested by a laboratory:
 - a) cataract, congenital glaucoma, congenital heart disease, hearing impairment, pigmentary retinopathy;
 - b) purpura, splenomegaly, microcephaly, developmental delay, meningocephalitis, radiolucent bone disease, jaundice that begins within 24 hours after birth..
- **Laboratory-confirmed CRS case:** An infant who is a suspected case (who has one condition from group (a) who meets the laboratory criteria for CRS case confirmation.
- **Congenital rubella infection (CRI):** An infant who does not have group (a) clinical signs of CRS but who meets the laboratory criteria for CRS is classified as having CRI.
- **Discarded:** A suspected case that does not meet the definition of clinically compatible case and has negative laboratory results after adequate testing.
- **Suspected Measles outbreak:** an occurrence of 3 or more suspected measles cases in one month in a rural ward/urban mahalla.
- **Non-measles non-rubella discarded case:** A suspect case that has been subjected to serology or virus identification but has not met any of the criteria for diagnosis, or a suspected case which has been linked epidemiologically to a confirmed case of another communicable disease.
- **Measles Incidence:** Annualized Incidence of Confirmed Measles (Laboratory and Epidemiologically confirmed) cases per million population.
Calculation: # of confirmed measles case ÷ Total population*10,00,000*(52/Epidemiological Week)
- **Discarded Rate :** Annualized Reporting rate of non-measles non-rubella cases at national level
Calculation: # of discarded case ÷ Total population*1,00,000*(52/Epidemiological Week)

Initiative of EPI Bangladesh

Key Advisors

Dr Md. Shamsul Haque (Line Director-MNC&AH)

Dr Mowla Baksh Chaudhury (PM-EPI) pmepi.dghs@gmail.com

Dr Rajendra Bohara, Team Lead, IVD, WHO bohara@who.int

Initiative supported by

Dr Balwinder Singh, MO, IVD WHO singhba@who.int

Dr Md. Tanvir Hossen, DPM-EPI & Surveillance tanvirantik@gmail.com

Dr Md. Monjurul Islam Fuad, DPM-Field Service drfuad.dmc@gmail.com

Dr Ariful Islam, Senior National Consultant-Measles Elimination, WHO IVD islam@who.int

Fozle Azim, Executive Assistant-Data Management , WHO IVD azimmd@who.int



Measles Rubella bulletin, Vol # 3, July 2021
Developed by WHO Bangladesh for EPI , DGHS .